

Fig. 7: An actual-size, single-side PCB for the controller board

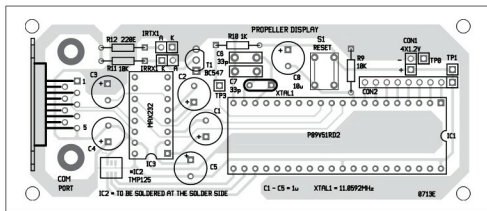


Fig. 8: Component layout for the PCB in Fig. 7

which means that in one whole minute it would be close to 1500 times. Hence this is the minimum effective rpm of the motor that has to be obtained after fitting the PCB and the components on the shaft.

To test the circuit, check test points for voltages as per the test point table. For functional testing, follow below-mentioned steps:

1. Connect the COM port of the computer to the display system (refer Fig. 3). You can also use a USB-to-serial converter

2. Place the interrupt sensor such that the beam is interrupted by the

beam interrupter

3. Open any hyper-terminal and type the first letter (all upper case)

4. Wait for change in the pattern of glowing LEDs

5. Enter the second letter and wait again for the changing pattern

6. Enter the remaining text similarly

7. When the text is complete, enter space and wait for the changing pattern

8. Press 'Enter.' All LEDs should glow

9. Wait until the LEDs turn off. Now start the motor. The text should

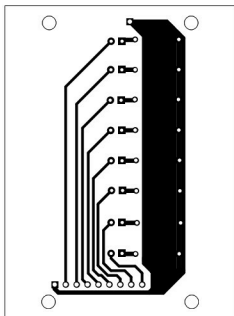


Fig. 9: An actual-size, single-side PCB for the LED strip

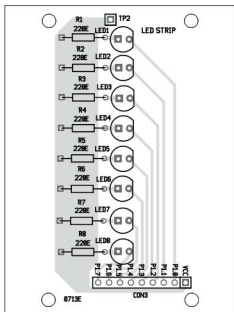


Fig. 10: Component layout for the PCB in Fig. 9

appear in some time on the circular display ●

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